

Xiwen Huang

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Education

Southern University of Science and Technology (SUSTech)

Shenzhen, China

Bachelor of Engineering in Information Engineering

Expected Jun.2027

GPA: 3.76/4.0, Honor Class

Analog Circuits Laboratory (100/100), Digital Circuits Laboratory (99/100), Fundamentals of Electric Circuits (95/100), Digital Signal Processing (95/100), Digital Circuits (92/100), Signals and Systems (92/100), Engineering Electromagnetics (90/100), Data communications and networking (90/100), Introduction to Computer Programming (96/100), Calculus II (94/100)

University of Pennsylvania

Philadelphia, USA

Exchange Program

Jan. 2026 - May 2026

GPA: 4.0 / 4.0

Machine Learning (A, 4.0/4.0), Foundations of AI: Deep Learning with Applications (A, 4.0/4.0), Musical Interfaces and Robotics (A+, 4.0/4.0)

Research Interest

I am an undergraduate student with a strong foundation in Robotics, EEE and AI. My research interests span generative models, reinforcement learning, and robotics, with a focus on building efficient AI systems with stronger generalization capabilities. I am particularly excited about embodied AI and the challenge of enabling agents to perceive, reason, and act in complex environments. I aim to bridge principled mathematical thinking with practical system design, and look forward to contributing to research at the frontier of these areas.

Honors & Awards

2026, 2024: SUSTech Study Abroad Scholarship

2025: Honorable Mention, Mathematical Contest in Modeling (MCM) — Team Leader

2024: SUSTech, Department of Electronics and Electrical Engineering - Wo Long Elite Class (Top 8%, Honor Class)

2024: Sole Winning Team, Nvidia Program of Blended Learning, MIT xPRO

2024: SUSTech Outstanding Student

2024: SUSTech Excellence Scholarship

Publication

CA-LRR: Controller-Aware Low-Rank Residual Refinement for Contact-Rich Aerial Manipulation

Second author; submitted to the IEEE International Conference on Robotics and Automation (ICRA 2027).

Research & Project Experience

CA-LRR for Contact-Rich Aerial Manipulation

Research Assistant; Westlake University; Advisor: Shiyu Zhao

Jun. 2026 - Sep. 2026

Contributed to implementing a low-rank residual RL framework for coordinated base-arm refinement around a frozen reference policy, using a five-dimensional SVD basis derived from successful simulation rollouts

Implemented controller-aware residual scaling based on tracking errors relative to prescribed bounds, with direction-preserving scaling to enforce action and joint limits

Engineered ROS-based hardware deployment integrating Vicon pose feedback, PreGME flight control, and reference and residual policy inference at 8 Hz

The resulting method increased mean task success from 44.91% to 56.72% across nine simulated load-slope conditions. Hardware experiments demonstrated pushing a cart with a total mass of 35.54 kg and, in a separate test, pushing across a 2 cm-high uneven surface

LLM Hallucination Research

Research Assistant; University of Pennsylvania; Advisor: Enric Boix; Ongoing

Feb. 2026 - May 2026

Fine-tuned TinyLlama-1.1B on a simple Q&A dataset using PEFT, implementing an abstention loss function that enabled the language model to output a learned <idk> token when uncertain, rather than generating answers that seemed plausible but actually incorrect (i.e., “hallucinations”).

Built on the theoretical framework of Kalai et al. (2025), which proves from a statistical perspective that binary evaluation metrics inherently incentivize language-model hallucination

Universal Vessel-like Structure Segmentation

Research Assistant; SUSTech; Advisor: Xiaoying Tang Jul. 2025 - Dec. 2025
Developed a universal vessel-like structure segmentation model based on nnU-Net incorporating prompt fusion and dual attention, achieving the segmentation of 10+ vessel-like structures across multiple modalities in a single network.
Improved average Dice from 84.92 to 85.79 (+0.87) and average cIDice from 89.58 to 91.38 (+1.80) on 7 2D datasets, outperforming single-task nnU-Net baselines

Learning Quadrupedal Robot Locomotion Policy with APG in Differentiable Simulation

Research Assistant; SUSTech; Advisor: Jiankun Wang Feb.2025 - Jun. 2025
Implemented differentiable-physics-based policy learning with MuJoCo XLA (MJX) and JAX, empirically demonstrating that this approach achieved better sample efficiency than PPO by training quadruped simulations and residual motion policies via analytical policy gradients
Adapted the ANYmal simulation environment to the Unitree AI platform; Built an interactive test visualization tool and implemented parameter saving/loading to ensure experiment reproducibility and fast deployment

Professional Experience

Shenzhen Guangxiang Technology Co., Ltd. Shenzhen, China
Intern Summer
Investigated collaborative robotic-arm applications for flexible LED display module production, mapping pick-place-alignment workflow across multi-SKU changeovers
Proposed a hybrid control scheme combining trajectory planning with RL-based local pose correction near targets, plus simulation training, deployment interfaces, and independent safety constraints

Shennan Circuits Co., Ltd Shenzhen, China
Intern, IT Process and Control Department Jan. 2025 - Feb. 2025
Led full-stack development of a shuttle bus waiting system with an HTML/CSS front-end and Java back-end, enabling 50+ employees to check real-time bus arrival times
Designed and managed an SQL database for bus routes, schedules, and real-time passenger waiting data

Skill Set

Programming: Python, C++, Java, HTML/CSS, SQL, LaTeX
Frameworks & Tools: PyTorch, PEFT, JAX/MJX, MuJoCo, ROS, PX4, Gazebo, nnU-Net, CLIP, SAM
Hardware & Embedded: Raspberry Pi, Embedded Systems, Circuit Design
Technical Focus: Machine Learning, Aerial Manipulation, Sim-to-Real, LLM Fine-tuning, Differentiable Simulation, AI Safety